Pranav Minasandra, Ph.D.

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Interests

I am a behavioural theorist and ethologist. My work explores the advantages of acknowledging the inherent complexity of behaviour in models, which sometimes leads to surprising insights. Broadly, I model behavioural transition dynamics across species. I also perform simulations to study how behaviour and cognitive abilities impact movement. Part of my research also concerns the use of more advanced programming methods in behavioural ecology.

Experience

Postdoc | **Max Planck Institute of Animal Behaviour, Konstanz, Germany** — 2024 - now
IMPRS-QBEE Postdoc Awardee.

Ph.D. | **Max Planck Institute of Animal Behaviour, Konstanz, Germany** — 2020 - 2024
Magna cum laude
Thesis: *"Exploring sequences of behaviour across scales and species."*
Also affiliated to: International Max Planck Research School of Quantitative Behaviour, Ecology, and Evolution; and the University of Konstanz.

M.Sc. | **Indian Institute of Science (IISc), Bengaluru, India** — 2019 - 2020
Master of Science in Biology.
Research at the Centre for Ecological Sciences, IISc.

B.Sc. (Research) | **Indian Institute of Science (IISc), Bengaluru, India** — 2015 - 2019
Major in Biology.
Research at the Centre for Ecological Sciences, IISc.

Publications

Peer-reviewed publications

1. Grout, E. M., Hass, C. C., Minasandra, P., Thomas, M., Ashbury, A. M., Ortega, J., Crofoot, M. C., Hirsch, B. T. & Strandburg-Peshkin, A. The vocal repertoire of white-nosed coatis: structural features, temporal dynamics, and social variations. *Bioacoustics*. 1–23. doi:10.1080/09524622.2026.2714161 (2026).
2. Gall, G. E., Demartsev, V., Minasandra, P., Baldoni, C., Cain, K. E. & Quinn, J. S. Examining Combinatoriality within the Pūkeko Vocal Repertoire. *Animal Behaviour*. 123431. doi:10.1016/j.anbehav.2025.123431 (2026).
3. Minasandra, P., Grout, E. M., Brock, K., Crofoot, M. C., Demartsev, V., Gersick, A. S., Hirsch, B. T., Holekamp, K. E., Johnson-Ulrich, L., Nayak, A., *et al.* Behavioral sequences across multiple animal species in the wild share common structural features. *PNAS* **122.** e2503962122. doi:10.1073/pnas.2503962122 (2025). [Best Paper Award | Press coverage].
4. DeNicola, V., Mezzini, S., Bursac, P., Minasandra, P. & Cagnacci, F. Effects of vasectomy on breeding-related movement and activity in free-ranging white-tailed deer. *Movement Ecology*. doi:10.1186/s40462-025-00554-5 (2025).
5. Grout, E. M., Ortega, J., Minasandra, P., Quin, M. J., Crofoot, M. C., Strandburg-Peshkin, A. & Hirsch, B. T. Whole Group Tracking Reveals That Relatedness Drives Consistent Subgrouping Patterns in White-Nosed Coatis. *Animal Behaviour* **216.** 175–193. ISSN: 0003-3472. doi:10.1016/j.anbehav.2024.08.010 (2024).
6. Minasandra, P., Jensen, F. H., Gersick, A. S., Holekamp, K. E., Strauss, E. D. & Strandburg-Peshkin, A. Accelerometer-based predictions of behaviour elucidate factors affecting the daily activity patterns of spotted hyenas. *Royal Society Open Science* **10.** 230750. doi:10.1098/rsos.230750 (2023).

7. Kunjar, S., Strandburg-Peshkin, A., Giese, H., Minasandra, P., Sarkar, S., Jolly, M. K. & Gradwohl, N. Link updating strategies influence consensus decisions as a function of the direction of communication. *Royal Society Open Science* **10.** 230215. doi:10.1098/rsos.230215.
8. Minasandra, P. & Isvaran, K. Truncated power-law distribution of group sizes in antelope. *Behaviour* **157.** 541–558. doi:10.1163/1568539X-bja10012 (2020).

Pre-prints and work in-prep

9. Minasandra, P. Anticipating others' future behaviours alters collective movement and enhances survival. *biorxiv* **122.** 2025–08. doi:10.1101/2025.08.14.670290 (2025). [In review].
10. Minasandra*, P., Sridhar*, V. H., Roche, D. G. & Planas-Sitjà, I. TracktorLive: An Integrated Real-Time Object Tracking and Response System. *bioRxiv*. 2026.03.12.711471. doi:10.64898/2026.03.12.711471 (2026). [In review, GitHub].
11. Minasandra, P., Demartsev, V., Johnson-Ulrich, L., Manser, M. & Strandburg-Peshkin, A. Spatiotemporal interplay of vigilance, movement, and collective behaviour in groups of meerkats. [Ch. 3 of Dissertation, available online].
12. Goldsborough, Z., Crofoot, M., Minasandra, P., León, J., Castillo-Caballero, P. L., del Rosario-Vargas, E. & Barrett, B. J. Habitual tool use on monopolizable resources affects group cohesion. doi:10.32942/X25H1Q (2025). [In review].

(* → co-first author)

Mentorship

Oakleigh Wilson (Ph.D. student) — 2026 - now
Role: External Advisor
Doctoral thesis chapter: Behaviour inference from accelerometer data through unsupervised machine learning.

Juee Dhar (Ph.D. student) — 2025 - now
Role: Mentor and Thesis Advisory Committee member.
Doctoral thesis: Sleep social dynamics in baboon groups.

Amlan Nayak (Master's student) — 2022 - 2023
Role: Mentor and primary co-advisor.
Master's thesis: Spatial variation of vigilance in meerkats.

Ananya Passi (Bachelor's student) — 2019
Role: Co-advisor
Summer research: Mathematical modelling of habitat use and population dynamics.

Awards and fellowships

ISBE Travel Award — 2026
Registration fees and travel costs to ISBE conference 2026. [≈1,600 €]

Best Paper Award—University of Konstanz — 2025
Chosen from among several nominations. [200 €]

Nominated—Schmidt Science Fellows programme — 2024
One of ten doctoral graduates nominated by the Germany-wide Max Planck Society.

IMPRS Postdoctoral Award — 2024 - 2025
6 months of pay at the scale of a postdoctoral researcher. [32,350 €]

ISBE Travel Award — 2024
2500 AUD [≈1,500 €].

DAAD—Graduate Student Scholarship Programme — 2020 - 2024
Nominated after evaluation of my proposal. [≈60,000 €]

Kishore Vaigyanik Protsahan Yojana — 2015 - 2020
Govt. of India scholarship for exceptional undergraduates in science. *All India Rank : 135*

Talks and posters

[Invited] International Conference of Neuroethology — 2026
Vancouver, Canada
"The Behavioural Observatory: Cross-species comparison of continuous decision-making"
(in the invited symposium *Brains in the Wild* by Jan Benda.)

International Society of Behavioural Ecology Meeting — 2026
Turin, Italy
"The Behavioural Observatory: Cross-species comparison of continuous decision-making"

[Invited] Talk at Laboratory for the Conservation of Endangered Species — 2025
Centre for Cellular and Molecular Biology, Hyderabad
"Three short stories in animal behaviour"

[Invited] CES Seminar Series — 2024
Indian Institute of Science, Bangalore
"Exploring sequences of behaviour in the wild across scales and species"

International Society of Behavioural Ecology Meeting — 2024
Melbourne, Australia
"Sequences of behaviour show a common statistical structure across species"

Gesselschaft für Primatologie Annual Conference — 2024
Konstanz, Germany
"How animals behave: Sequences of behavior show a common statistical structure across species"

Descriptive & Normative Models of Collective Behaviour — 2022
School of Mathematics, University of Leeds
"Looking into hyena daily activity patterns using accelerometers"

Teaching

Introduction to Git — ongoing
Instructor and organiser: Hands on in-house workshop. [link to class notes]

Introduction to remote code servers like GitHub — ongoing
Instructor and organiser: Hands on in-house workshop. [link to class notes]

LaTeX for academic documents — ongoing
Instructor and organiser: Hands on in-house workshop. [link to class notes]

Quantitative Ecology: Research Design and Inference — 2019
Teaching assistant: Taught coding in R, conducted several classes, graded assignments, managed course website.

Technical skills

Programming languages and related
- Python, Julia, R, Bash, LaTeX, and C.
- HTML, CSS, and JavaScript.
- Version control with Git; GitHub (including actions, workflows, and organisation management).
- Tools for Linux server setup and management.
- Compiled and hosted a Linux from Scratch OS.

Artificial Intelligence (AI)
- Agentic code generation and validation.
- Hosting local Large Language Models (LLMs); data inference with LLMs (with `ollama`); vision-capable LLMs and video processing;
- Design of basic agentic loops.

Other specialised computational skills
Machine learning; Parallel processing and Inter-Process Communication; Agent-based models; Analysis of complex multivariate time-series data.

Mathematical modelling
Dynamical (PDE) models incorporating spatial and stochastic variables; Probability models; Numerical simulations of all the above.

Other academic services

Reviewing
Animal Biotelemetry; Ecology and Evolution; Methods in Ecology and Evolution; PLOS One; Integrative and Comparative Biology; Animal Behaviour.

Representation 2024 - 2025
Postdoc representative at the Max Planck Institute of Animal Behaviour.

References

Dr Ariana Strandburg-Peshkin,
Max Planck Institute of Animal Behavior
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Dr Kavita Isvaran,
Indian Institute of Science
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Dr Andrew Berdahl,
University of Washington
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Prof Meg Crofoot,
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